The Ephemeris

June 2023
Volume 34 Number 02- The Official Publication of the San Jose Astronomical Association

June 2023 - September 2023 Events
Schedule subject to change
Please check our website for up to date information.

Board & General Meetings
Saturday 6/3, 7/1, 8/5,9/2
Board Meetings: 6 -7:30pm
General Meetings: 8-9:30pm

Solar Sunday (2-4pm)
Sunday 6/4, 7/9, 8/6, 9/10

Intro to the Night Sky Class &
Houge Park 3Q In-Town Star Party
Saturday 6/16, 7/14, 8/18, 9/8

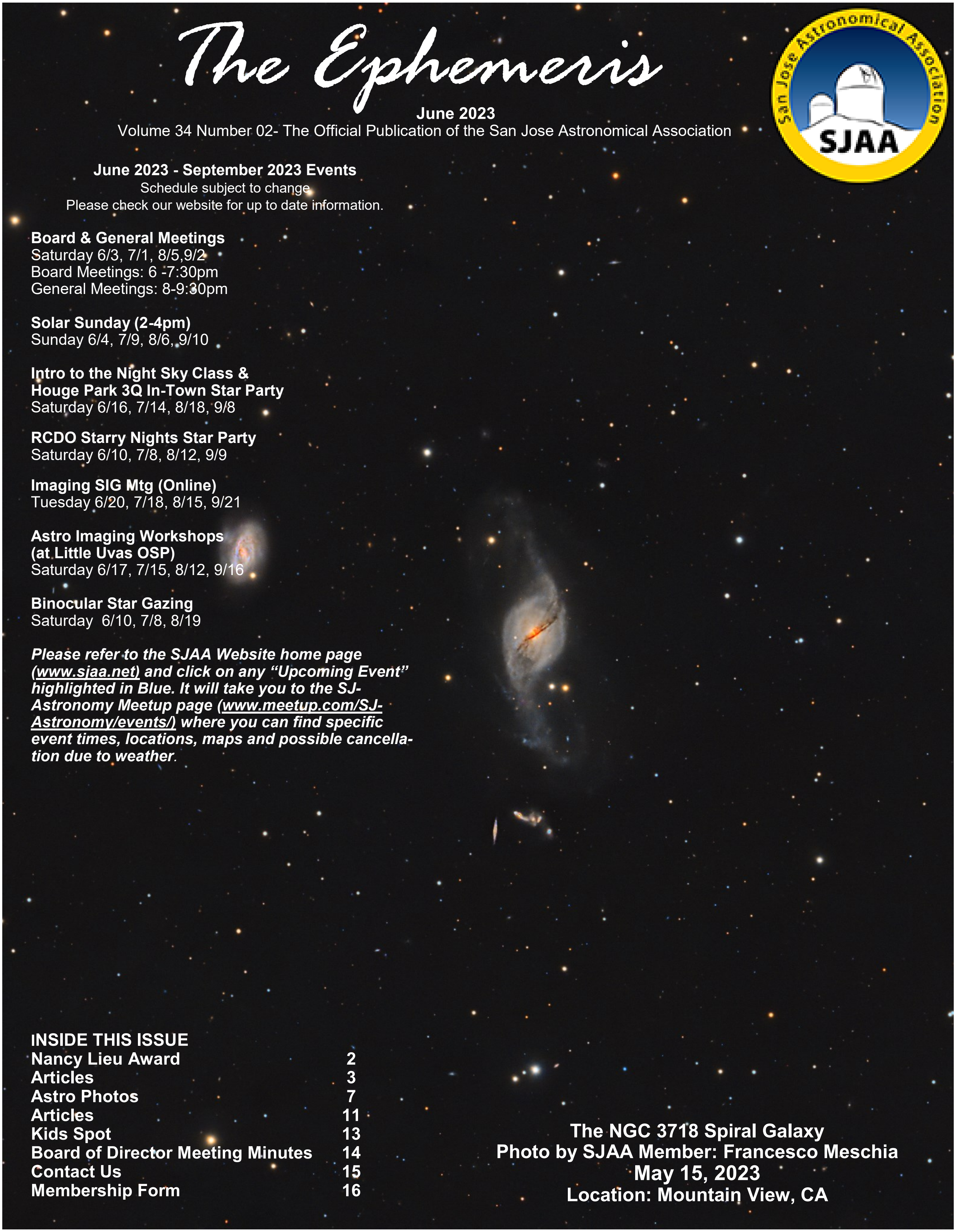
RCDO Starry Nights Star Party
Saturday 6/10, 7/8, 8/12, 9/9

Imaging SIG Mtg (Online)
Tuesday 6/20, 7/18, 8/15, 9/21

Astro Imaging Workshops
(at Little Uvas OSP)
Saturday 6/17, 7/15, 8/12, 9/16

Binocular Star Gazing
Saturday 6/10, 7/8, 8/19

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.



INSIDE THIS ISSUE

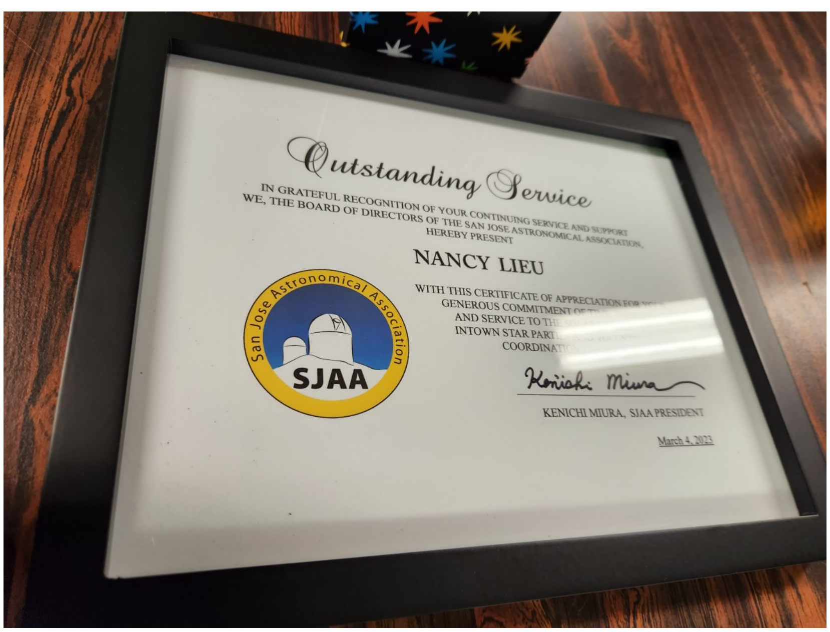
Nancy Lieu Award	2
Articles	3
Astro Photos	7
Articles	11
Kids Spot	13
Board of Director Meeting Minutes	14
Contact Us	15
Membership Form	16

The NGC 3718 Spiral Galaxy
Photo by SJAA Member: Francesco Meschia
May 15, 2023
Location: Mountain View, CA

Excellence Award for Nancy Lieu

Credit: Wolf Witt

In 2018, Nancy proactively joined SJAA's solar astronomy crew to help with events during solar minimum, when solar astronomy volunteers are few and far between. Since then, she's been a steady and invaluable presence during solar events in Houge Park and at remote locations, and she's played an integral part in training additional solar telescope operators. Furthermore, in 2022, Nancy stepped up to cover SJAA's Volunteer Coordinator role as the prior person withdrew unexpectedly, and she's been a key contributor at other events like ITSP.



The SJAA award for outstanding service presented to Nancy Lieu by former President Ken Miura on 6th May 2023

SJAA Logo Wear
Credit: Marianne Damon

Show our pride and let the public know who we are! Making your own custom logo-wear is easy and quick. Jon Anderson recently made some baseball caps for his School Star Party Program and they look great! I wore my long-sleeved shirt at the District 9 Music in the Valley Festival and people knew who I was for sure! It's both an identifier and conversation starter. We have a webpage on our SJAA website where you can:

- Download the SJAA logo file to your computer.
- Select a logo-wear company. SJAA does not endorse any one company. You can use a company you have ordered from before, or find one that has the items you want. We've listed some of the companies that you can find online and that have received favorable reviews.
- Select the "Design/Create Your Own" option
- Select the product available for that option
- Follow the instructions for importing the SJAA logo file onto the site
- Follow the instructions for placing the logo on the front & back, sizing the logo, and inserting a text box if you wish to add your name. You can customize the font type and color of text within the text box.
- Save the design, or finish and place in your cart. When finished with all items, go to cart and check out/order.

I've road-tested this method and within one week I received my long-sleeved shirt with a pocket logo on the front, where I inserted my first name, and a large logo on the backside. I also made a carry-bag, with the logo and my full name on one side, and just the logo on the other side. Some companies have basic, but adequate, selections of colors and styles, and others have more extensive options. For example, a company might have basic shirts, sweatshirts and baseball caps, and another company might have shirts of different brands, more colors, a variety of hats, and a selection of jackets. You can select the type, color, quality and pricing of your choosing.

We are looking forward to seeing more of our logo at club events! I remember a young boy and his mom I walked over to in the parking lot. They were wondering if they were at the right place for one of our meetings: "Mom, she's wearing an Orion Nebula t-shirt, I think we're in the right place"…had I been wearing my SJAA shirt, there would have been no doubt at all!

On our SJAA website, look under Resources for SJAA Logo-Wear. If you are reading this online, here is the link to the logo-wear webpage: SJAA Logo-Wear - SJAA

Observation Report: The Box

Credit: Joe Fragola

At the conclusion of the Starry Nights star party on August 15, I was finally able to check out a group of galaxies nicknamed “The Box”. This galaxy cluster had been on my target list for observing sessions at Rancho Canada del Oro earlier in the week. I didn’t get to view it previously, so I was anxious to give it a try on Saturday night after the star party wrapped up.

The four galaxies, all classified as spirals, are located in the constellation Coma Berenices. They are members of a galaxy cluster designated as Hickson 61.

Hickson 61 is a tight grouping of four galaxies that form a nearly perfect rectangle. Contained within a field of just 3.8', they all will fit into the same medium-power field.

Here is the info for the four galaxies that are part of “The Box” (magnitude data is from SkySafari Pro).

NGC 4173, mag. 13.6 (upper left)
NGC 4169, mag. 12.2 (upper right)
NGC 4175, mag. 13.2 (lower left)
NGC 4174, mag. 14.3 (lower right)

Stars Don’t Twinkle, & ‘Wrong way’ Planets?

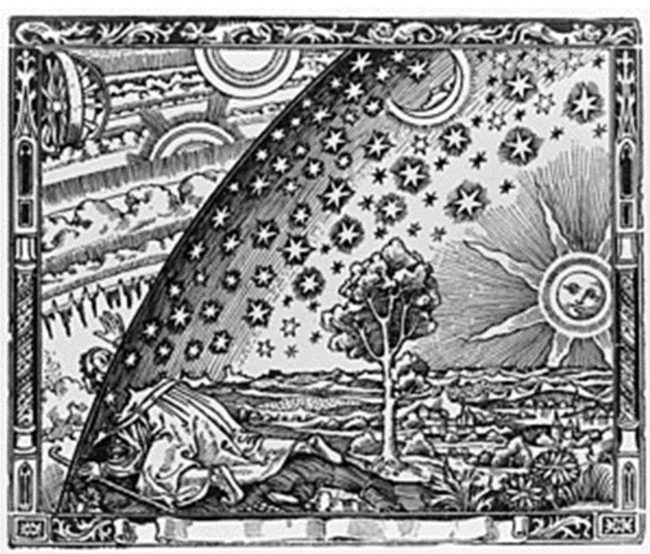
By: Jordan Makower

A person steps out into the cold, clear night, and gazes up into the sky. Bright, flickering lights appear and become more numerous. You may think that all those lights are stars. Stars appear to ‘twinkle’ to us on Earth, but not when viewed from space. Our shifting atmospheric layers distort the light we receive from stars, but not from our Sun. It is apparently too large to be greatly affected by the clear atmosphere, but the tiny rays of light that reach us from the more distant stars are distorted, and they seem to flicker.

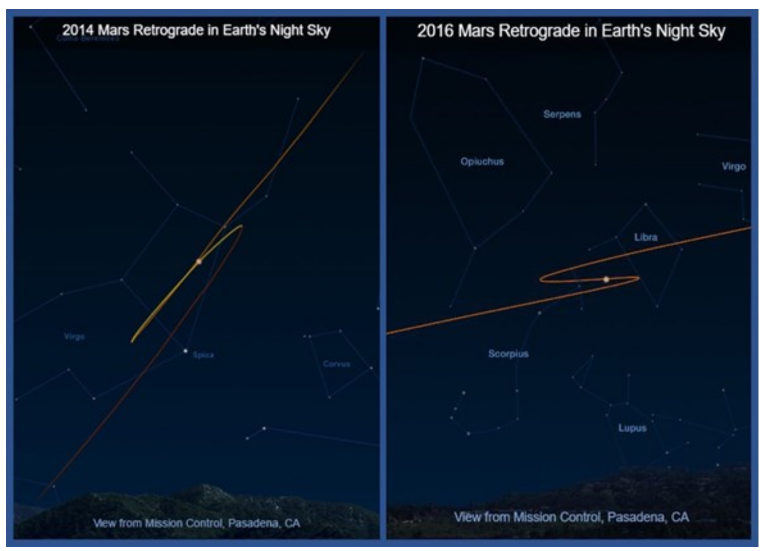
Usually, that’s how we can tell a star from a planet at night. Planets shine by reflecting the light from our Sun but are closer than the stars, and their light *normally* isn’t affected by out atmosphere, except when they are rising or setting. At those times, we are looking through a larger ‘chunk’ of air, and then even planets may twinkle!

The word ‘planet’ is derived from a Greek word meaning ‘wanderer’ since, over a period of time, planets slowly change their position among the stars, moving through constellations. During our lifetimes, the stars maintain their distances from other stars, and we have assigned groups of them to remind us of ancient legends, naming them as ‘constellations’. Different cultures may have other groupings and names for them, but the International Astronomical Association only lists 88 constellations.

As our world turns from West to East, the stars seem to move, getting higher in the East and lower in the West, and the ‘outer planets’ seem to drift through the constellations, faster than the stars ‘move’, in the same direction … *usually*! Centuries ago, people kept to the illusion that stars were fixed onto a huge dome-sphere, which slowly moved around Earth.



Credit: Wikipedia.org. Flammarion engraving



Credit: mars.nasa.gov

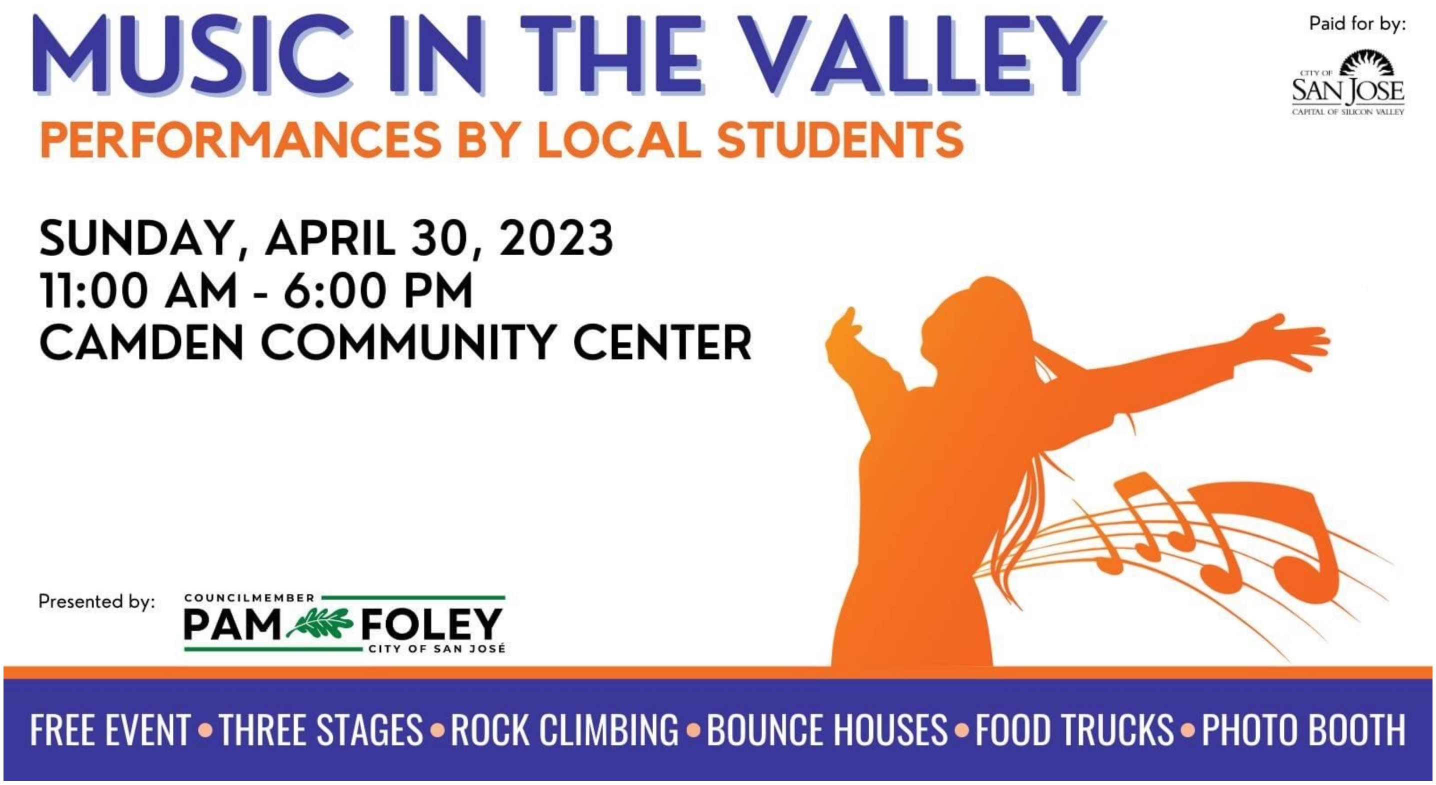
To account for movements of the Sun, Moon, and Planets, different transparent ‘spheres’ were proposed, and objects on them all got higher in the East and lower in the West (the Geocentric Model of Ptolemy). Difficulties with this model arose when Mars, Jupiter, and Saturn (easily visible objects) would seemingly slow down, stop, back up, and then move forward again (See NASA diagram, above). Each of these did this at different rates and times! These problems baffled people before Galileo came along with his telescope.

What did he see that challenged the ideas of the spheres? How would the ‘wrong-way’ motion of the planets be explained? What things can you see with your binoculars/telescope to disprove Ptolemy? How would those observations do that?

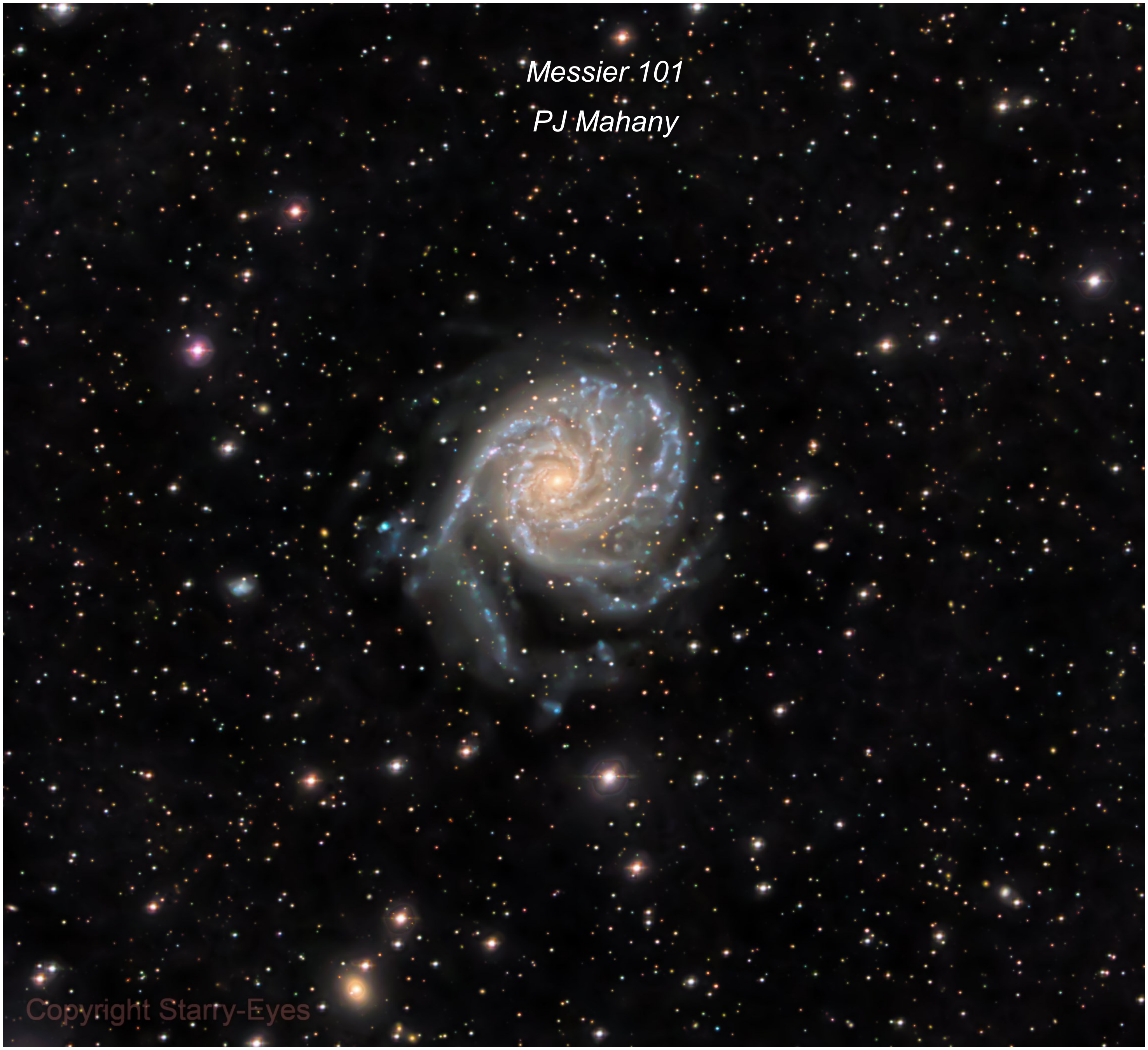
District Nine's Music in the Valley Event

Credit: Marianne Damon

Thanks to an agreement with the City of San Jose, SJAA uses Houge Park's Building 1 as our homebase. Houge Park is administered by Parks, Recreation, Neighborhood Services (PRNS), and falls in San Jose's District 9. Pam Foley is the District 9 councilmember. She was first elected in 2018 and was just re- elected in January of 2023. District 9 has a history of hosting a community festival each year at Camden Community Center, and we have had a booth at most of these events. This year the festival underwent a revision and had a new theme of showcasing the district's middle and high school music programs. The event was held on Sunday, April 30. Two outdoor and one indoor stage hosted continuous performances. Rather than having many booths sponsored by businesses as in the past, the focus this year was on community services. There were food trucks, free popsicles from SJPD, some fun lawn games, and a very tall climbing wall which required harnesses. SJAA was invited to have a booth, and our group was headed up by (as usual) Rob Jaworski. Kanch and Jon Anderson helped out in the morning, and Marianne provided solar views with her Coronado PST. For times when clouds covered the sun, we had the UV solar bead bracelet activity for the younger children. Of interest, Rob's son played with his school on one of the stages. Rob and Marianne packed up around 4:30, just as a small group of high-school musicians were launching into a heavy metal session. The crowd was definitely lighter than in prior years, but we connected with lots of interested and interesting people. We for sure let District 9 and PRNS staff know that we were present and active with our community outreach.



SJAA Member Astrophoto Gallery



Date: 13 May 2023

Location: Sierra Nevada Mountains, Near Yosemite, CA

Imaging Telescope: Celestron RASA 11”

Imaging Camera: QHY268c

Mount: Software Bisque Paramount MX+ GEM

Software: AstroSharp Ltd SharpCap , Open PHD Guiding Project PHD2, Skyhound SkyTools , Software Bisque

SJAA Member Astrophoto Gallery

Dates: May 17, 2023

Location: Home

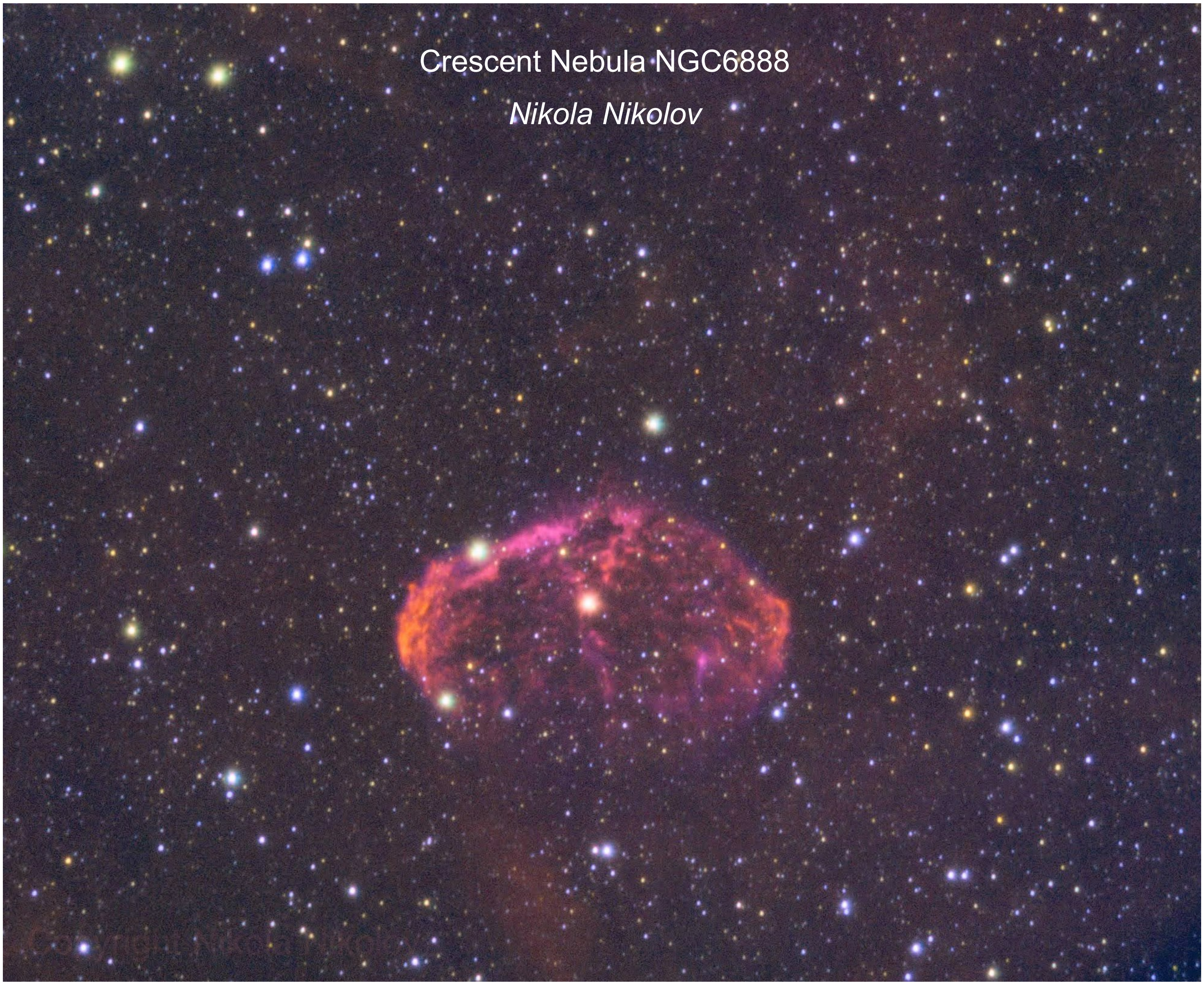
Telescope: Celestron C6

Camera: ZWO ASI533MC Pro

Mount: Celestron Advance VX

Software: Adobe Photoshop, Pleiades Astrophoto PixInsight

SJAA Member Astrophoto Gallery



Dates: April 13, 2023, April 14, 2023, April 20, 2023, April 26, 2023, May 15, 2023, May 20, 2023, May 22, 2023

Location: San Jose, CA

Telescope: William Optics ZenithStar 61 / ZS61

Camera: ZWO ASI83MC Pro

Mount: Orion Sirius EQ-G .

Software: Adobe Photoshop, Aries Productions Astro Pixel Processor (APP), Pleiades Astrophoto PixInsight, Stefan Berg Nighttime Imaging 'N' Astronomy (N.I.N.A. / NINA)

SJAA Member Astrophoto Gallery

Dates: April 21 2023

Location: Home, Silicon Valley, California

Telescope: GSO 10" f/8 Ritchey-Chretien Truss Tube

Camera: ZWO ASI1600MM Pro

Mount: Astro-Physics 1100GTO

Software: Adobe Lightroom Classic, Adobe Photoshop, KDEdu KStars, Pleiades Astrophoto PixInsight

NASA's Perseverance Rover Captures View of Mars' Belva Crater

Credit: NASA

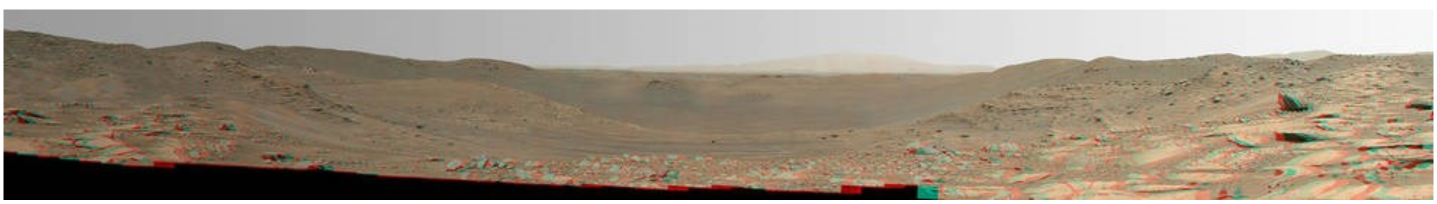
The 152 images that make up this mosaic of Belva Crater were taken by the Mastcam-Z instrument aboard NASA's Perseverance Mars rover on April 22, 2023, the 772nd Martian day, or sol, of the mission. Belva is a 0.6-mile-wide (0.9-kilometer wide) impact crater within the much larger Jezero Crater.

Credits: NASA/JPL-Caltech/ASU/MSS

The Mastcam-Z instrument aboard NASA's Perseverance Mars rover recently collected 152 images while looking deep into Belva Crater, a large impact crater within the far larger Jezero Crater. Stitched into a dramatic mosaic, the results are not only eye-catching, but also provide the rover's science team some deep insights into the interior of Jezero.

"Mars rover missions usually end up exploring bedrock in small, flat exposures in the immediate workspace of the rover," said Katie Stack Morgan, deputy project scientist of Perseverance at NASA's Jet Propulsion Laboratory in Southern California. "That's why our science team was so keen to image and study Belva. Impact craters can offer grand views and vertical cuts that provide important clues to the origin of these rocks with a perspective and at a scale that we don't usually experience."

On Earth, geology professors often take their students to visit highway "roadcuts" –places where construction crews have sliced vertically into the rock to make way for roads – that allow them to view rock layers and other geological fea-



This anaglyph of Perseverance's mosaic of Belva Crater can best be viewed with red-blue 3D glasses.

Credits: NASA/JPL-Caltech/ASU/MSS

tures not visible at the surface. On Mars, impact craters like Belva can provide a type of natural roadcut.

Signs of Past Water

Perseverance took the images of the basin on April 22 (the 772nd Martian day, or sol, of the mission) while parked just west of Belva Crater's rim on a light-toned rocky outcrop the mission's science team calls "Echo Creek." Created by a meteorite impact eons ago, the approximately 0.6-mile-wide (0.9-kilometer-wide) crater reveals multiple locations of exposed bedrock as well as a region where sedimentary layers angle steeply downward.

These "dipping beds" could indicate the presence of a large Martian sandbar, made of sediment, that billions of years ago was deposited by a river channel flowing into the lake that Jezero Crater once held.

More About the Mission

A key objective for Perseverance's mission on Mars is astrobiology, including caching samples that may contain signs of ancient microbial life. The rover will characterize the planet's geology and past climate, pave the way for human exploration of the Red Planet, and be the first mission to collect and cache Martian rock and regolith.

Subsequent NASA missions, in cooperation with ESA, would send spacecraft to Mars to collect these sealed samples from the surface and return them to Earth for in-depth analysis.

The Mars 2020 Perseverance mission is part of NASA's Moon to Mars exploration approach, which includes Artemis missions to the Moon that will help prepare for human exploration of the Red Planet.

NASA Pursues Lunar Terrain Vehicle Services for Artemis Missions

Credits: NASA

Artist's concept of NASA's next-generation Lunar Terrain Vehicle on the surface of the Moon.

Credits: NASA

NASA is seeking industry proposals for a next-generation LTV (Lunar Terrain Vehicle) that will allow astronauts to go farther and conduct more science than ever before as they explore the south polar region of the Moon during Artemis missions.

Artemis astronauts will drive to explore and sample more of the lunar surface using the LTV than they could on foot. NASA will contract LTV as a service from industry rather than owning the rover. Contracting services from industry partners allows NASA to leverage commercial innovation and provide the best value to U.S. taxpayers while achieving its human spaceflight scientific and exploration goals.

“We want to leverage industry’s knowledge and innovation, combined with NASA’s history of successfully operating rovers, to make the best possible surface rover for our astronaut crews and scientific researchers,” said Lara Kearney, manager of NASA’s Extravehicular Activity and Human Surface Mobility program at the agency’s Johnson Space Center in Houston.

The LTV will function like a cross between an Apollo-style lunar rover and a Mars-style uncrewed rover. It will support phases driven by astronauts and phases as an uncrewed mobile science exploration platform, similar to NASA’s Curiosity and Perseverance Mars rovers. This will enable continued performance of science even when crews are not present on the lunar surface. Artemis astronauts will use the LTV to traverse the lunar surface and transport scientific equipment, extending the distances they can cover on each moonwalk.

Under the Lunar Terrain Vehicle Services request for proposals, NASA has provided requirements for companies interested in developing and demonstrating the LTV, including an approach that encourages companies to produce an innovative rover for use by NASA and other commercial customers for multiple years.

NASA intends to use the LTV for crewed operations beginning with Artemis V in 2029. Prior to crew arrival, the rover will be used for uncrewed and commercial activities once it lands on the lunar surface.

Through Artemis, NASA will send astronauts – including the first woman and first person of color – to explore the Moon for scientific discovery, economic benefits, and to build the foundation for crewed missions to Mars. Together, NASA’s Space Launch System rocket, Orion spacecraft, Gateway lunar orbital outpost, advanced spacesuits and rovers, and human landing systems are the agency’s foundation for deep space exploration.

Kid Spot Jokes:

- **Why did Mickey Mouse go to outer space?**
  To find Pluto.

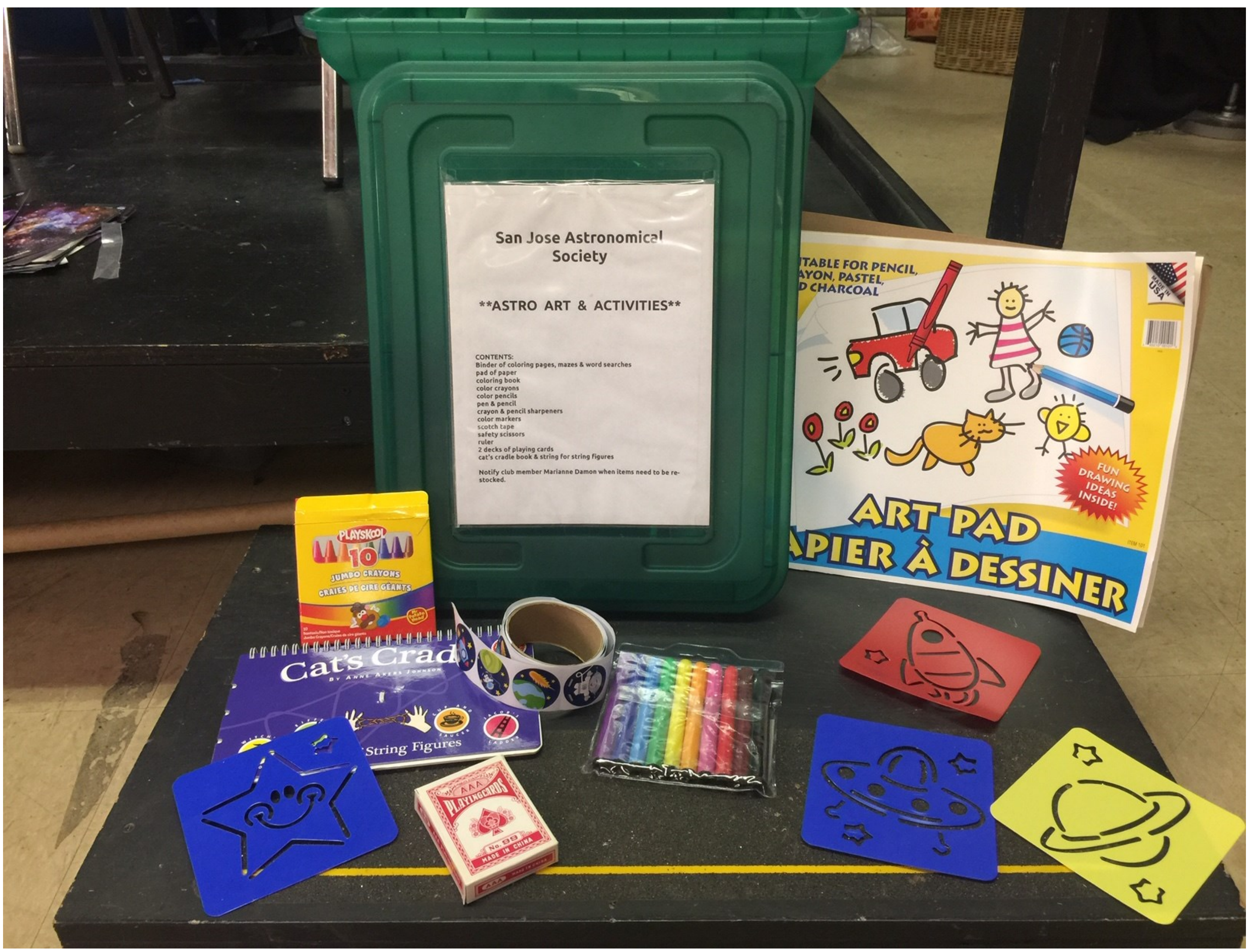
- **Why didn't the sun go to college?**
  Because it already had a million degrees!

- **What did the doctor say to the rocketship?**
  Time to get your booster shot!

Children's Activity Box

Credit: Marianne Damon

Picture this scenario: You need to work on some equipment in Building 1 but you also have your five-year-old with you. You've already been to the playground but now it's time for you to go inside and get down to business. How do you keep your child occupied? We have an activity box! Inside the large storage bin is a big pad of drawing paper, markers, colored pencils, coloring sheets (with astronomy themes, of course!), and a few fun activities. You can find the bin on the bottom shelf of our main cabinet next to the library cabinet.



From the Board of Directors

Announcements

Current SJAA Members can benefit from signing up for the “SJ Astronomy” group on Meetup (https://www.meetup.com/sj-astronomy/) as well as the “SJAA Observers” Google group (https://groups.google.com/g/sjaa-observers).

Open Positions: Librarian, Intro To The Night Sky

Board Meeting Excerpts

April 8, 2023

In attendance
Vini, Aroon, Swami, Rob, Peter, Glenn, Chris, Ken, Wolf
Absences: Sukhada
Guests: Marianne, Emi, Leslie, Joe, Kaushal, Sunny, Michael, Songchun, Jonathan, Renuka

Hotspot at Houge Park
SJAA will pay for hotspot at Houge Park, $110 every month, and reimburse Aroon for $37.
Glenn made the motion to approve the payment above, Peter seconded. All approved, no abstentions or objections. Motion carried.

Temporary exclusive use of HP building for the Youth Shakespeare group
Rob to email the program leads with the relevant info, ask them for a schedule if they need access to HP during the dates of Saturday, May 13th to Sunday, May 28th.

Reimbursements
Rob made a motion to reimburse the costs incurred by Marianne as specified above, Vini seconded. All approved, no abstentions or objections. Motion carried.

Rob made a motion to reimburse the costs incurred by Marianne, for a ladder for the 24” Dob, Vini seconded. All approved, no abstentions or objections. Motion carried.

Peter made the motion to approve the $99 expense for electionbuddy for conducting the board elections in February, Vini seconded. All approved, no abstentions or objections. Motion carried.

May 6, 2023

In attendance
Vini, Aroon, Ken, Wolf, Swami, Peter, Sukhada, Chris,
Absences: Rob, Glenn
Guests: Emi, Jonathan, Nancy, Neeti, Joe, Michael, Marianne, Kanch

Excellence Award 2023
Ken presented the award to Nancy who was recommended by Wolf. The certificate was accompanied by a gift - a coffee mug from the National Science Museum in Tokyo. For a photo of Nancy Lieu with the award and mug please see Page 2 of this issue.

Minors policy
The working team of Wolf, Marianne, and Rob will continue working on the Minors policy and bring it back to the board for re-approval and adoption. Aroon will follow up on the code of conduct and determine how to get signatures from the leaders, directors, and officers.

USB microphone for board meetings
Vini made a motion to spend up to $200 to buy Focusrite Scarlett 2i2 computer interface that will allow USB microphones to be added to anybody’s laptop for the board meeting. The same can also be used for the speaker talks as needed. Sukhada seconded. All approved, no abstentions or objections. Motion carried.

June 3, 2023

In attendance
Swami, Wolf, Vini, Glenn, Rob, Aroon, Chris
Absences: Ken, Sukhada, Peter
Guests: Emi, Hytham

24” Dob Task Force
The task force set up the 24” Dob, successfully aligned it, used it to observe a few objects, and has concluded that the scope is usable. Relevant equipment, and a suitable storage location for the 24” Dob is currently being identified.

Minors Policy
The working team of Wolf, Marianne and Peter will clean up the Minor's Policy document. The Code of Conduct document will be moved to the Policy book. Aroon will reach out to members for legal expertise.

Additional Zoom license(s) for Program leaders

Glenn made a motion to approve the funding for a third Zoom license to be used for SIG meetings, Wolf seconded. All approved, no abstentions or objections. Motion carried.

SJAA Library

SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

http://www.sjaa.net/programs/fixit/

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

http://www.meetup.com/SJ-Astronomy/

Quick STARt Program

The Quick STARt Program, headed up by Jonathan Lawton, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick STARt program is undergoing a redesign to make it more accessible to SJAA members, so signups are currently on hold.

http://www.sjaa.net/programs/quick-start/

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

http://www.sjaa.net/programs/beginners-astronomy/

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

http://www.sjaa.net/programs/loaner-telescope-program/

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

http://www.sjaa.net/programs/imaging-sig/

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in Astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

http://www.sjaa.net/programs/school-star-party/

SJAA Contacts

President/Director:	Aroon Nair
Vice-President:	Chris Gottbrath
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Swami Nigam
Director:	Vini Carter
Director:	Wolf Witt
Director:	Ken Miura
Director:	Peter Melhus
Director:	Glenn Newell
Director:	Sukhada Palav
Ephemeris Newsletter - Content Editor:	Abishek Ramasubramanian
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Hy Murveit
Intro to the Night Sky:	Vacant
In-Town Star Party:	Muditha Kanchana
Library:	Vacant
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Beth Johnson
Quick STARt	Jonathan Lawton
Donations:	Rich Gregor
Solar:	Wolf Witt
Starry Nights:	Joe Fragola
School Events:	Jon Anderson
Refreshments:	Marianne Damon
Social Media Manager:	Beth Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki

E-mails: http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
http://www.sjaa.net/contact

Place
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here

San Jose Astronomical Association
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San Jose, CA 95159-8243

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **$20** Regular Membership with **online** Ephemeris

☐ **$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: **http://www.sjaa.net/sjaa-newsletter-ephemeris/**

Questions? Send e-mail to: **memberships@sjaa.net**

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: **http://www.sjaa.net/join-the-sjaa/**

Name: ______________________________

Address: ______________________________

City/ST/Zip: ______________________________

Phone: ______________________________

E-mail address: ______________________________